

Sharp scalar counterexamples for three extended inexact proximal algorithms

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Abstract

Reem and Reich prove that three inexact proximal algorithms remain well defined beyond the parameter regimes of the original convergence theorems, and ask whether the generated sequences still converge to a zero. All three answers are negative already on the real line for the maximally monotone identity operator. At the boundary relative-error value one, explicit admissible choices give

$$a_n = 1 - \frac{1}{(n+2)^2}, \quad x_n = \prod_{j=0}^{n-1} a_j = \frac{n+2}{2(n+1)} \longrightarrow \frac{1}{2},$$

although the unique zero is 0. Beyond the boundary, constant choices give geometric divergence. The examples cover Algorithms 9.1, 10.1, and 11.1 of the source, including parameters satisfying its formula (10.1).

1 The three source questions

Reem and Reich study inexact resolvent inclusions in a reflexive Banach space and prove well-definedness of many implicit algorithmic schemes [1]. Three of their extensions are followed by explicit convergence questions.

For the Solodov–Svaiter scheme, the source says:

that any such sequence converges weakly to a zero of A . Under further assumptions it was shown in [89, Theorem 2.4] that these sequences converge strongly to a zero of A .

The following theorem shows that Algorithm 9.1 is well defined even if $\sigma \geq 1$ (it is, however, an open problem whether the generated sequence converges to a zero of A , since the analysis in [89] depends on the assumption that $\sigma \in [0, 1)$). Moreover, if $\sigma > 0$, then either x_n is a zero of A or (9.1) has strictly inexact solutions.

Thus Question A is whether Algorithm 9.1 converges when $\sigma \geq 1$.

For Iusem–Pennanen–Svaiter, the exact source text is:

based on this lemma; however, in [56, Method 1] A is assumed to be maximally monotone; while this implies that A^{-1} is maximally monotone and thus it is also maximally ρ -hypomonotone for all arbitrary small $\rho > 0$, the exact value of ρ to be used in ν from (10.1) is not mentioned.) Anyway, Theorem 10.2 below shows that ν can be an arbitrary nonnegative number, independently of σ and ρ , and usually there is some freedom in the value of the inexact solution (y_n, η_n) . As a result, if, in particular, we want ν to be defined by (10.1), then we can take any $\sigma \in [0, \infty)$ and any $\rho \in [0, \widehat{\lambda}/2]$. It is, however, an open problem whether the sequence $(x_n)_{n=0}^\infty$ converges weakly to a zero of A when ν is not assumed to satisfy (10.1) or when it satisfies (10.1) but $\sigma \geq 1$, since the convergence analysis in [56] depends on (10.1) and also on the assumption that $\sigma \in [0, 1)$.

Question B asks about arbitrary ν , and also about ν defined by formula (10.1) when $\sigma \geq 1$. Finally, for the variable-metric Parente–Lotito–Solodov scheme:

shown in [71, Theorem 4.2] that $(x_n)_{n=0}^\infty$ converges to a zero of A . Under additional assumptions, a rate of convergence was established [71, Theorem 4.4].

The following theorem shows that if no enlargements are allowed, then Algorithm 11.1 is well defined for all $\sigma \in (0, \infty]$ (including $\sigma \geq 1$), all $\theta \in \mathbb{R}$ (if $\theta < 0$, then we interpret $[1-\theta, 1+\theta]$ as the set $\{t \in \mathbb{R} : 1+\theta \leq t \leq 1-\theta\}$), all $c_n > 0$, $n \in \mathbb{N} \cup \{0\}$ (not necessarily bounded away from zero by some $c > 0$), and all positive definite linear operators $M_n : X \rightarrow X$, $n \in \mathbb{N} \cup \{0\}$ (without any restriction on their eigenvalues). (It is, however, an open problem whether the generated sequence converges to a zero of A in this extended version since the convergence analysis in [71] depends on the assumptions imposed in Algorithm 11.1.) In addition, if $\sigma_n > 0$, then either x_n is a zero of A or (11.2) has strictly inexact solutions.

Question C asks whether Algorithm 11.1 converges under the extended well-definedness hypotheses of Theorem 11.2. We answer all three questions negatively.

2 A common scalar model

Let $X = \mathbb{R}$ with its usual Hilbert norm and let

$$A(t) = t \quad (t \in \mathbb{R}).$$

This A is maximally monotone and has the unique zero 0. Put

$$a_n = 1 - \frac{1}{(n+2)^2} \quad (n \geq 0), \quad x_0 = 1. \quad (1)$$

The elementary telescoping identity

$$\prod_{j=0}^{n-1} a_j = \prod_{k=2}^{n+1} \frac{(k-1)(k+1)}{k^2} = \frac{n+2}{2(n+1)} \quad (2)$$

will force every boundary example to converge to 1/2 rather than to the zero of A .

3 Solodov–Svaiter: failure at and above one

In the notation of Algorithm 9.1, one step must satisfy

$$\xi_n \in A(y_n), \quad (3)$$

$$0 = \xi_n + \mu_n(y_n - x_n) + \eta_n, \quad (4)$$

$$|\eta_n| \leq \sigma \max\{|\xi_n|, \mu_n|y_n - x_n|\}, \quad (5)$$

and, unless $\xi_n = 0$ or $y_n = x_n$, the update is

$$x_{n+1} = x_n - \frac{\xi_n(x_n - y_n)}{\xi_n^2} \xi_n. \quad (6)$$

Theorem 1. *Question A has a negative answer.*

1. At $\sigma = 1$ there is an infinite admissible sequence converging to $1/2 \neq 0$.
2. For every $\sigma > 1$ there is an admissible sequence with $|x_n| \rightarrow \infty$ geometrically.
3. At $\sigma = 1$ the algorithm may even satisfy its stopping test at an arbitrary nonzero point.

Proof. Set $\mu_n = 1$ and, for the first assertion, define

$$y_n = \xi_n = a_n x_n, \quad \eta_n = (1 - 2a_n)x_n. \quad (7)$$

Relations (3) and (4) are identities. Since $1/2 < a_n < 1$ and $x_n > 0$,

$$|\eta_n| = (2a_n - 1)x_n \leq a_n x_n = \max\{|\xi_n|, |y_n - x_n|\}.$$

Thus (5) holds with $\sigma = 1$. Neither stopping condition occurs, and (6) gives $x_{n+1} = y_n = a_n x_n$. Formula (2) proves $x_n \rightarrow 1/2$.

Now fix $\sigma > 1$. Choose a constant $a > 1$ such that

$$2a - 1 \leq \sigma a. \quad (8)$$

If $1 < \sigma < 2$, any $1 < a \leq (2 - \sigma)^{-1}$ works; if $\sigma \geq 2$, every $a > 1$ works. Replacing a_n by a in (7) satisfies all three conditions, and (6) gives $x_n = a^n x_0$.

Finally, at a prescribed $x \neq 0$, take $y = x$, $\xi = x$, and $\eta = -x$. Then (3)–(5) hold at $\sigma = 1$, but the rule stops because $y = x$, although $0 \notin A(x)$. $\square$

4 Iusem–Pennanen–Svaiter

With $\lambda_n = 1$ and $Z = \mathbb{R}$, Algorithm 10.1 requires

$$\eta_n \in A(y_n) + y_n - x_n, \quad |\eta_n| \leq \nu |y_n - x_n|, \quad x_{n+1} = y_n - \eta_n. \quad (9)$$

Theorem 2. *Question B has a negative answer. At $\nu = 1$ there is an admissible sequence converging to $1/2 \neq 0$, and for every $\nu > 1$ there is an admissible geometrically divergent sequence. Moreover, $\nu = 1$ arises from source formula (10.1) with $\hat{\lambda} = 1$, $\rho = 1/6$, and $\sigma = 3$.*

Proof. At the boundary define

$$y_n = (1 - a_n)x_n, \quad \eta_n = (1 - 2a_n)x_n. \quad (10)$$

Because $A(y_n) = y_n$, the inclusion in (9) is equality. Furthermore,

$$|\eta_n| = (2a_n - 1)x_n \leq a_n x_n = |y_n - x_n|, \quad y_n - \eta_n = a_n x_n.$$

Equations (2) and (9) give the claimed nonzero limit. For $\nu > 1$, choose a constant $a > 1$ satisfying $2a - 1 \leq \nu a$ as in (8), and use (10) with $a_n = a$. Then $x_{n+1} = ax_n$.

It remains to match the formula branch of the source question. Formula (10.1) is

$$\nu = \frac{\sqrt{\sigma + (1 - \sigma)(2\rho/\hat{\lambda})^2 - 2\rho/\hat{\lambda}}}{1 + 2\rho/\hat{\lambda}}.$$

For $(\hat{\lambda}, \rho, \sigma) = (1, 1/6, 3)$, the square root is $5/3$, and hence $\nu = (5/3 - 1/3)/(1 + 1/3) = 1$ exactly. $\square$

5 Parente–Lotito–Solodov

Specialize Algorithm 11.1 to $c_n = 1$, $M_n = I$, $\epsilon_n = 0$, and $\tau_n = 1$ (allowed, for example, with $\theta = 1/2$). Its step conditions and update become

$$\xi_n \in A(y_n), \quad (11)$$

$$\eta_n = \xi_n + y_n - x_n, \quad (12)$$

$$\eta_n^2 \leq \sigma_n^2 (\xi_n^2 + (y_n - x_n)^2), \quad (13)$$

$$x_{n+1} = x_n - \frac{\xi_n(x_n - y_n)}{\xi_n^2} \xi_n. \quad (14)$$

Theorem 3. *Question C has a negative answer.*

1. *With global parameter $\sigma = 1$, one can have every $0 < \sigma_n < 1$ and $\sup_n \sigma_n = 1$, while $x_n \rightarrow 1/2 \neq 0$.*
2. *For every global $\sigma > 1$, admissible constant parameters can make $|x_n| \rightarrow \infty$ geometrically.*

Proof. For the first assertion set

$$y_n = \xi_n = a_n x_n, \quad \eta_n = (2a_n - 1)x_n, \quad \sigma_n = \frac{2a_n - 1}{\sqrt{a_n^2 + (1 - a_n)^2}}. \quad (15)$$

Equations (11)–(13) hold, the last with equality. Since $1/2 < a_n < 1$,

$$(2a_n - 1)^2 < a_n^2 + (1 - a_n)^2,$$

so $0 < \sigma_n < 1$. As $a_n \rightarrow 1$, also $\sigma_n \rightarrow 1$, and therefore the global choice $\sigma = 1$ has $\sigma_n \in [0, \sigma)$ as required by the extended theorem. The update (14) is $x_{n+1} = a_n x_n$, and (2) applies.

For $a > 1$, put

$$q(a) = \frac{2a - 1}{\sqrt{a^2 + (a - 1)^2}}.$$

We have $q(1) = 1$ and q is continuous. Given global $\sigma > 1$, choose s with $1 < s < \min\{\sigma, \sqrt{2}\}$ and then choose $a > 1$ sufficiently close to one that $q(a) < s$. Use (15) with constant a and constant $\sigma_n = s$. All extended conditions hold and $x_n = a^n x_0$. $\square$

6 Interpretation and verification

The three examples share one mechanism. Classical convergence assumes a uniform relative-error gap below one. At the boundary, each scalar update may still be a strict contraction, but the deficits $1 - a_n$ can be summable; then the infinite product is positive. Above the boundary, expansion factors are admissible. Thus the strict uniform gap is not a cosmetic proof artifact.

The accompanying script `verify_counterexamples.py` checks the first 100 iterations in exact rational arithmetic, including every inclusion, relative-error inequality, update identity, the closed form (2), and the formula-(10.1) parameter calculation. Since all spaces are one-dimensional, weak and norm convergence coincide.

Targeted searches of the exact source phrases, the original three algorithm papers, the run registry, and current proximal-method literature found convergence results under strict relative-error conditions, but no prior statement of these boundary examples. The counterexamples are elementary; literature novelty and the transcription of the three algorithms nevertheless deserve specialist review.

Limitations. These examples answer the universal extended-convergence questions negatively. They do not classify every individual extended parameter choice, nor do they contradict the original convergence theorems, all of which impose the strict conditions that the examples deliberately violate.

Human review recommendation. Verify equations (3)–(6), (9), and (11)–(14) directly against Algorithms 9.1, 10.1, and 11.1 of the source. If confirmed, notify the authors: the three questions have sharp, one-dimensional negative answers.

References

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- [2] M. V. Solodov and B. F. Svaiter, *A hybrid approximate extragradient–proximal point algorithm using the enlargement of a maximal monotone operator*, Set-Valued Anal. 7 (1999), 323–345.
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